

02-719 Week 2

Genomics and Epigenetics of the Brain

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Basic Experiment Setup

Suppose we are running an experiment with 3 different animals, in two conditions. Let A , B , C be in condition A and D , E , F be in condition B. Let a scale from 0 to 10 define the amount of open chromatin. Suppose we run our experiment, and we get:

Species x Condition	Open Chromatin
A	2
B	4
C	3
D	7
E	10
F	8

From this, we can then compare the effects on the two different conditions on epigenetics. Some statistics we can use to quantify the difference are

- Mean and Std. Dev
- t-test
- Linear Regression
- etc.

Now, if today we want to understand the conditions' effects on multiple genes, we can create matrix where there are g rows, where g is the number of genes we care about, and c number of columns, where c is the number of species x conditions. Data can then be compared and processed efficiently using a computer program.

Introduction to R Programming Language

- Very similar to Matlab
- Handles matrix operations very well
 - For loops are quite slow
- R is open source
 - The underlying code is available
 - No license required
- Many people have put together statistical and genomic packages in for R
 - [Bioconductor](#) - Open source software for bioinformatics
 - Limma - linear models for microarray (and RNA-Seq)

Data / Class Types in R

- Boolean, Integer, Numeric (double), String, Factor
- Vector - 1D, every element same data type
- List - 1D, every element can be different type
- Matrix - 2D, every element same data type
- DataFrame - 2D, every column same data type
- Factor - Used for representing qualitative variables in statistical operations

Assigning Different Types

- `x <- 1`: assignment operator
- `class(x)`: identify data type with class function
- `x <- c(1, 2, 3)`: assign a vector
- `x <- matrix(c(1, 2, 3, 4, 5, 6), 2, 3)`: assign a matrix. This example makes a matrix of 3 rows, 2 cols. Numbers fill rows first (1, 2, 3) in col 1.
- `f <- data.frame(num=c(1, 2, 3, 4), letter=c("a", "b", "c", "d"))`: sets data in a data frame. column name is the key.
- `rownames(f) <- c("first", "second", "third", "fourth")`: sets row names for a dataframe

Gene Ontology

- When we perform a ChIP-Seq experiment, have peaks (signal) that can potentially cover any part of the entire genome, not just genes.
- The GREAT gene ontology analysis tool is designed to handle this case.

Expected Counts

Problem: We have a genome that contains 100 genes.

- 10 of those genes are involved in DNA replication
- We conduct an experiment where key replication gene is knocked out
- As a result, 20 genes change in expression
- 8 of those genes are involved in replication
- What is the significance?

To solve: Let n be number of genes (20), k be number of genes in category (8), and p be probability that any given gene is in the category: ($10/100 = 0.1$).

By simple probability, we would expect that ($20 \text{ genes} \times 0.1 \text{ probability} =$) 2 genes are in the category. What we observe was 8 genes, which is much higher than the predicted 2 genes. $P_{\text{binom}} = 0.0004$.

Binomial P-value

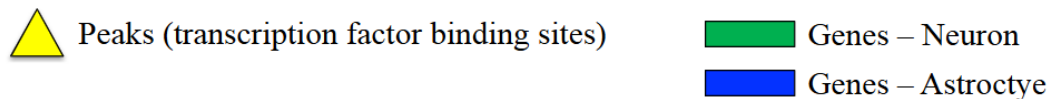
- We can do statistical enrichments based on groups of genes:
 - How many genes do I see in a category?
 - How many genes would I expect by chance?

$$\Pr(k; n, p) = \binom{n}{k} \cdot p^k \cdot (1-p)^{n-k}$$

- In the formula above,
 - n = number of trials (number of genes)
 - k = number of successes (number of genes in category)
 - p = probability of success (probability that any given gene is in the category)

Mapping Regulatory Elements to Genes

The assumptions of the binomial test may not be valid.



Example 1:



Peaks tend to be near neuron marker genes. Works well!

Example 2:



Peaks tend to be near neuron genes. Works well?

NO: Genes may not be evenly distributed.

GREAT

Solution: Partition the genome based on the gene ontology category of the nearest gene. Then, adjust the probability of a hit based on the proportion of the genome that is covered, not the total number of genes in the category.

Example 2:



Original Test: P-value = $\text{binom}(n=10, k=9, p=0.5) = 0.01$

New Test: P-value = $\text{binom}(n=10, k=9, p=0.9) = 0.73$

[FILL friday lecture]

Human Accelerated Regions (HARS)

What regions are highly constrained across vertebrates, but show differences in the human lineage (usually relative to chimpanzee)?

- Reporter Assay: what patterns of gene expression are driven by a given sequence (draw)?
- HARs tend to be developmental enhancers.
- High Throughput Reporter Assay: testing thousands of sequences in cultured neurons reveals differences between human and chimpanzee sequence. Most differences lead to higher neural activity (enhancers) in humans.

Linking Genotype to Phenotype

Can we relate the rate of evolution “correlate” with enhancer activity?

- Can we use linear regression to compare two continuous variables across species (compare body size with evolutionary rate across mammals?)
- Phylogenetic regression controls for evolutionary relationships

Relative Evolutionary Rates - RER Coverage

Does the rate of evolution “correlate” with enhancer activity?

1. Align a set of sequences together (amino acid or nucleotide)
2. Calculate the rate of evolution along each branch in the tree
3. Compare the branches in which a behavior evolved to other branches
4. Check for a significant association (faster or slower) evolution along the highlighted branches

[Fill]

Example: Vocal Learning

Vocal learning is the ability to modify acoustic output based on auditory experience, enabling to imitate and produce new sounds.

- Vocal learning has undergone convergent evolution, having evolved independently in multiple lineages of mammals and birds.

Modules of Vocal Learning

If an animal falls under all three modules, it is considered a vocal learner.

- **Vocal Versatility**
 - Acoustic repertoire complexity
 - Source-filter model of production
 - Pitch shifting / pitch matching
 - Vocal imitation / mimicry
 - Improvisation / innovation
- **Vocal Production Variability**

- Ontogenetic / developmental variability
- Babbling / subsong / plastic song
- Critical periods for vocal plasticity
- Closed- and open-ended learning

- **Vocal Coordination**

- Antiphonal calling
- Jamming avoidance
- Chorusing
- Turn-taking
- Duetting
- Rhythmic entrainment

Molecular Brain Similarities

[Fill image]

Vocal Learning Enhancer Convergent Evolution

- Most enhancers shows lower predicted activity in vocal learning species.
- Enhancers show a tendency to be near autism-associated genes.